

**BANARAS HINDU UNIVERSITY**

B.Voc. Semester II Examination, 2021-22

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**Paper: BVCA-201 — Introduction To Databases**

Time: Three Hours

Maximum Marks: 70

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*Instructions: Attempt all sections. Candidates are required to write their answers in their own words as far as practicable. Symbols have their usual meanings.*

**PART — A**

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**Short Answer Questions** (Answer all parts)*[10 × 2 = 20]*

1. Define the following database terms:

- (a) Data model
- (b) Database schema
- (c) Client/server architecture
- (d) Owner entity type
- (e) Weak entity type
- (f) Entity
- (g) Attribute value
- (h) Relation state
- (i) Degree of a relation
- (j) Normalization

**PART — B**

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**Medium Answer Questions** (Attempt all questions; either/or type)*[5 × 4 = 20]*

2. What is the difference between logical data independence and physical data independence? Which one is harder to achieve, and why?

**OR**

Discuss the different data types supported in SQL.

3. Discuss the entity integrity and referential integrity constraints. Why is each considered important?

**OR**

Discuss Functional Dependency with a suitable illustrative example.

4. What is a relationship type? Explain the differences among a relationship instance, a relationship type, and a relationship set.

**OR**

What is a participation role? When is it necessary to use role names in the description of relationship types?

5. What is meant by a recursive relationship type? Give some examples of recursive relationship types.

**OR**

What is SQL? Write down the syntax to create a table and insert records into a table.

6. Why are duplicate tuples not allowed in a relation?

**OR**

What is the difference between a key and a super key?

## **PART — C**

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**Long Answer Questions** (Answer any three questions)

[3 × 10 = 30]

7. Describe the three-schema architecture. Why do we need mappings between schema levels? How do different schema definition languages support this architecture?
8. Discuss the different types of user-friendly interfaces and types of users who typically use each.
9. Discuss the various reasons that lead to the occurrence of NULL values in relations.
10. Discuss the role of a high-level data model in the database design process.